Polar Form, De Moivre's Theorem & Roots of Unity

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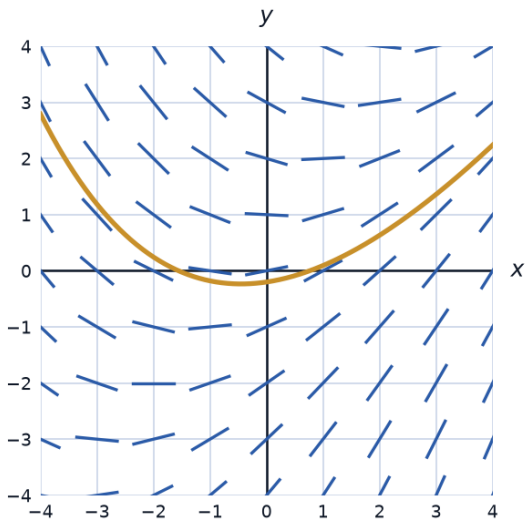
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Polar and Euler Forms

Use De Moivre's theorem to prove that $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$. Hence solve $4x^3 - 3x = 0$ on $x \in [-1, 1]$.

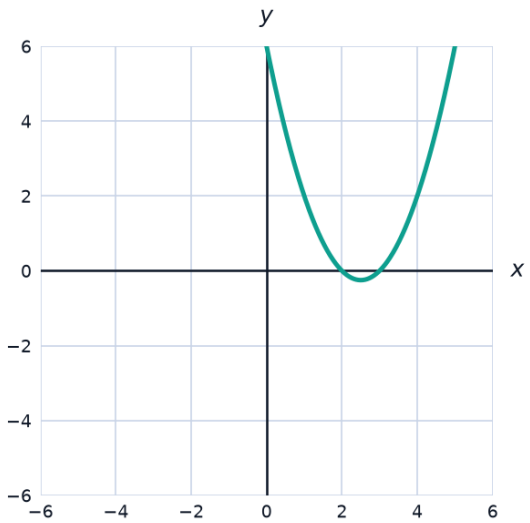
2 [Foundational] • Target: De Moivre's Theorem for Powers

(IB AA/AI HL) Given the initial value problem $\frac{dy}{dx} = x + y$ with $y(0) = 1$, use Euler's numerical method with step size $h = 0.1$ to approximate $y(0.2)$. Compare your numerical result with the exact analytical solution.



3 [Foundational] • Target: Roots of Complex Numbers

Consider the function $f(x) = x^2 - 5x + 6$. The graph of this function is a parabola. Determine the x-coordinates of the points where the graph intersects the x-axis.



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4 [Foundational] • Target: Roots of Unity & Factorization

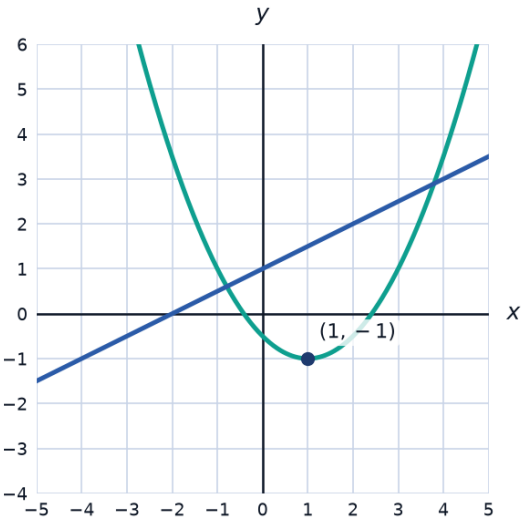
Consider the function $f(x) = ax^2 + bx + c$. The graph of $f(x)$ is a parabola that passes through the points $(-1, 12)$, $(1, 2)$, and $(2, 9)$. Find the value of the parameter b.

5 [Foundational] • Target: Polar and Euler Forms

(Paper 3 Extended Investigation) Consider the recursive Babylonian sequence defined by $x_0 = 3$ and $x_{n+1} = \frac{1}{2}\left(x_n + \frac{5}{x_n}\right)$ for $n \geq 0$. (a) Use the AM-GM inequality to prove that $x_n \geq \sqrt{5}$ for all $n \geq 1$. (b) Show that the sequence is strictly decreasing for $n \geq 1$, and deduce that $\lim_{n\to\infty} x_n = \sqrt{5}$. (c) Prove the quadratic error relationship $|x_{n+1} - \sqrt{5}| = \frac{(x_n - \sqrt{5})^2}{2x_n}$.

6 [Foundational] • Target: De Moivre's Theorem for Powers

Two straight paths in 3D space are modeled by lines $L_1 : \mathbf{r}_1 = (1; 2; 0) + \lambda(1; -1; 2)$ and $L_2 : \mathbf{r}_2 = (0; 1; -1) + \mu(2; 1; 1)$. Show that L_1 and L_2 are skew, and find the shortest perpendicular distance between them.



7 [Foundational] • Target: Roots of Complex Numbers

Consider the quadratic equation $x^2 - kx + 8 = 0$, where k is a real parameter. The graph of the corresponding quadratic function $y = x^2 - kx + 8$ is a parabola. Given that the parabola intersects the x-axis at exactly one point, find the value of k.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Roots of Unity & Factorization

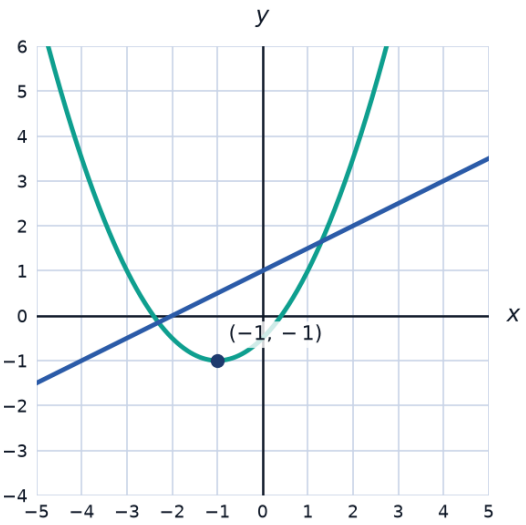
(Paper 3 Extended Investigation) Define the integral family $I_n = \int_0^1 x^n e^x \, dx$ for integers $n \geq 0$. (a) Evaluate I_0. (b) Use integration by parts to establish the reduction formula $I_n = e - nI_{n-1}$ for $n \geq 1$. (c) Hence determine the exact value of $\int_0^1 x^3 e^x \, dx$. (d) Prove that $0 < I_n < \frac{e}{n+1}$, and deduce $\lim_{n \to \infty} I_n$.

9 [Intermediate] • Target: Polar and Euler Forms

(IB AI HL) A market transition matrix $T = (0.7, 0.2; 0.3, 0.8)$ represents weekly customer movement between Brand A and Brand B. Find: (a) the probability that a customer initially with Brand A is with Brand B after 2 weeks; (b) the long-term steady-state market distribution.

10 [Intermediate] • Target: De Moivre's Theorem for Powers

Prove by mathematical induction that $5^n - 1$ is divisible by 4 for all positive integers $n \in \mathbb{Z}^+$.



11 [Intermediate] • Target: Roots of Complex Numbers

Determine the equation of the oblique asymptote to the graph of $f(x) = \frac{2x^2 - 3x + 4}{x - 1}$.

12 [Intermediate] • Target: Roots of Unity & Factorization

Use trigonometric substitution to evaluate the indefinite integral $\int \frac{1}{(4 + x^2)^{3/2}} \, dx$.

13 [Intermediate] • Target: Polar and Euler Forms

(IB AI HL) A communications network with 6 nodes $\{A, B, C, D, E, F\}$ has weighted edges: AB=4, AC=2, BC=1, BD=5, CD=8, CE=10, DE=2, DF=6, EF=3. Use Kruskal's algorithm to determine the minimum spanning tree, stating the order of edges added, and find the total tree weight.

14 [Intermediate] • Target: De Moivre's Theorem for Powers

Consider the function $f(x) = \frac{\sin(2x)}{\cos(x)}$. For the domain $x \in [0, 2\pi]$ such that $\cos(x) eq 0$, find the exact value of the definite integral $\int_0^{\pi/2} f(x) dx$.

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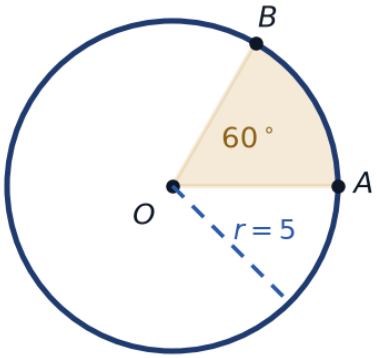
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging] • Target: Roots of Complex Numbers • Cross-Lesson: 6.03 Markov Chains (AI HL)**

Let $\mathbf{a} = (2;\ -1; 3)$ and $\mathbf{b} = (1; 4;\ -2)$. Find the cross product $\mathbf{a} \times \mathbf{b}$, and determine the exact area of the triangle with adjacent sides $\mathbf{a}$ and $\mathbf{b}$.

16 **[Challenging] • Target: Roots of Unity & Factorization • Cross-Lesson: 1.01 Proof by Mathematical Induction**

Using the binomial series for $(1 - x^2)^{-1/2}$, derive the Maclaurin series expansion for $f(x) = \arcsin x$ up to and including the x^5 term. State its radius of convergence.



17 **[Challenging] • Target: Polar and Euler Forms • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra**

A survey was conducted on the number of hours students spent studying per week. The results are summarized in the histogram below. The total number of students surveyed is 120. Let x represent the number of students who studied for more than 3 hours but at most 4 hours. Find the value of x.

18 **[Challenging] • Target: De Moivre's Theorem for Powers • Cross-Lesson: 1.04 Partial Fractions**

Decompose the rational expression $\frac{5x^2 - 3x + 2}{(x - 1)(x^2 + 1)}$ into partial fractions.

19 **[Challenging] • Target: Roots of Complex Numbers • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)**

The derivative of the function $f(x) = 3x^2 - 5x + 2$ is given by $f'(x)$. Find the value of $f'(2)$.

20 **[Challenging] • Target: Roots of Unity & Factorization • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem**

A random variable X follows a binomial distribution $B(n, p)$. The number of trials is $n = 10$ and the probability of success in a single trial is p. If the mean of X is 4, find the variance of X.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Polar and Euler Forms

Express $z = -2 - 2\sqrt{3}i$ in Euler's exponential form $re^{i\theta}$ with $\theta \in (-\pi, \pi]$.

a. $4e^{-i2\pi/3}$

b. $4e^{i4\pi/3}$

c. $2e^{-i2\pi/3}$

d. $4e^{-i\pi/3}$

22 [Foundational] • Target: Polar and Euler Forms

Convert $z = 6e^{i3\pi/4}$ into exact Cartesian form $x + iy$.

a. $-3\sqrt{2} + 3\sqrt{2}i$

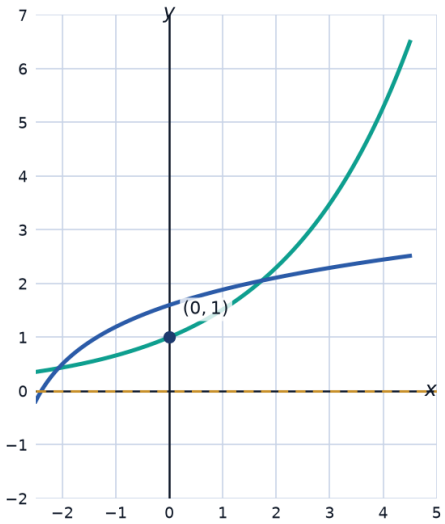
b. $3\sqrt{2} - 3\sqrt{2}i$

c. $-3 + 3\sqrt{3}i$

d. $-3\sqrt{2} - 3\sqrt{2}i$

23 [Foundational] • Target: Polar and Euler Forms

If $z_1 = 3e^{i\pi/3}$ and $z_2 = 2e^{i\pi/6}$, evaluate the quotient $\frac{z_1}{z_2}$ in exponential form.



a. $\frac{3}{2}e^{i\pi/6}$

b. $\frac{3}{2}e^{i\pi/2}$

c. $6e^{i\pi/6}$

d. $1.5e^{-i\pi/6}$

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24 [Foundational] • Target: De Moivre's Theorem for Powers

Use De Moivre's Theorem to evaluate $(1 + i\sqrt{3})^6$.

a. 64

b. -64

c. $64i$

d. $-64i$

25 [Foundational] • Target: De Moivre's Theorem for Powers

Evaluate $(\cos(\pi/5) + i\sin(\pi/5))^{10}$.

a. 1

b. -1

c. i

d. $-i$

26 [Foundational] • Target: De Moivre's Theorem for Powers

Express $\cos(3\theta)$ in terms of $\cos\theta$ using De Moivre's Theorem $(\cos\theta + i\sin\theta)^3$.

a. $4\cos^3\theta - 3\cos\theta$

b. $3\cos^3\theta - 4\cos\theta$

c. $\cos^3\theta - 3\cos\theta$

d. $4\cos^3\theta + 3\cos\theta$

27 [Foundational] • Target: Roots of Complex Numbers

Find all cube roots of $8i$ in polar form.

a. $2e^{i\pi/6}, 2e^{i5\pi/6}, 2e^{-i\pi/2}$

b. $2e^{i\pi/3}, 2e^{i\pi}, 2e^{i5\pi/3}$

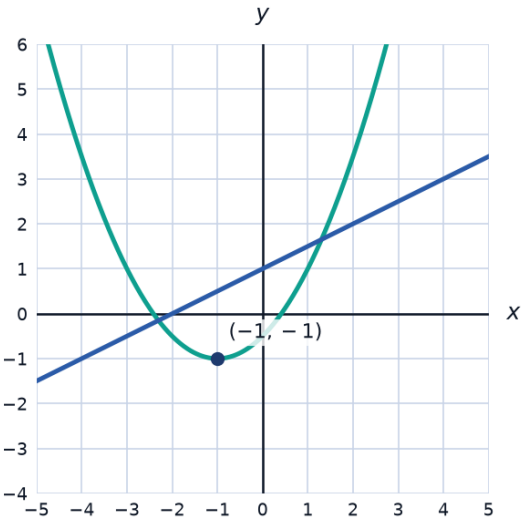
c. $8e^{i\pi/6}, 8e^{i5\pi/6}, 8e^{-i\pi/2}$

d. $2,\ -1 + i\sqrt{3},\ -1 - i\sqrt{3}$

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28 [Foundational] • Target: Roots of Complex Numbers

How are the n distinct nth roots of any complex number $z = re^{i\theta}$ distributed on the Argand plane?



a. They lie on a circle of radius $r^{1/n}$ centered at the origin, evenly spaced by $\frac{2\pi}{n}$ radians

b. They lie along a straight line through the origin

c. They form an ellipse with eccentricity $1/n$

d. They cluster exclusively in Quadrant I

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Roots of Complex Numbers

What is the sum of all n distinct nth roots of any complex number z ($n \geq 2$)?

a. 0

b. 1

c. n

d. z

30 [Intermediate] • Target: Roots of Unity & Factorization

If $\omega = e^{i2\pi/3}$ is a primitive complex cube root of unity, what is the value of $1 + \omega + \omega^2$?

a. 0

b. 1

c. -1

d. 3

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31 [Intermediate] • Target: Roots of Unity & Factorization

Factor $z^4 + 1$ into two real quadratic factors.

a. $(z^2 - \sqrt{2}z + 1)(z^2 + \sqrt{2}z + 1)$

b. $(z^2 + 1)^2$

c. $(z^2 - 1)(z^2 + 1)$

d. $(z^2 - 2z + 1)(z^2 + 2z + 1)$

32 [Intermediate] • Target: Roots of Unity & Factorization

The four 4th roots of unity ($z^4 = 1$) form the vertices of which geometric shape on the Argand plane?

a. A square inscribed in the unit circle with vertices at $\pm 1, \pm i$

b. An equilateral triangle

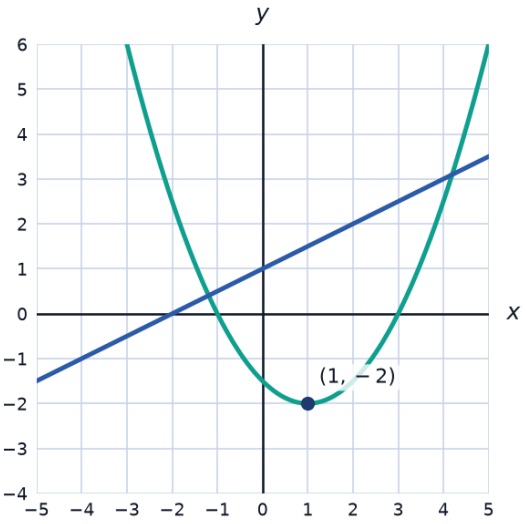
c. A regular hexagon

d. A rectangle of perimeter 4

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33 **[Intermediate]** • **Target: Polar and Euler Forms**

Find all cube roots of unity.



a. $1,\ \frac{-1+i\sqrt{3}}{2},\ \frac{-1-i\sqrt{3}}{2}$

b. $1, i, -1$

c. $1, -1, i$

d. $1, i\sqrt{3}, -i\sqrt{3}$

34 **[Intermediate]** • **Target: De Moivre's Theorem for Powers**

Consider the complex numbers $z_1 = 5 - 2i$ and $z_2 = 1 + 4i$. Find the value of $z_1 - z_2$ in the form $x + yi$.

a. 4 - 6i

b. 6 + 6i

c. 4 + 6i

d. 6 - 6i

35 **[Intermediate]** • **Target: Roots of Complex Numbers**

Simplify $\frac{19-9i}{-3+5i}$, giving your answer in the form $x + yi$.

a. $-3 - 2i$

b. -4

c. -1

d. -5

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36 [Intermediate] • Target: Roots of Unity & Factorization

Simplify $(4 + 2i)(-5 - 5i)$, giving your answer in the form $x + yi$.

a. $-6 - 33i$

b. $-8 - 33i$

c. $-12 - 33i$

d. $-10 - 30i$

37 [Intermediate] • Target: Polar and Euler Forms

Simplify $\frac{12 - 14i}{4 + 2i}$, giving your answer in the form $x + yi$.

a. $-3i$

b. $-6i$

c. $1 - 4i$

d. $-2i$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: De Moivre's Theorem for Powers • Cross-Lesson: 1.04 Partial Fractions

Simplify $\frac{18 + i}{4 + 3i}$, giving your answer in the form $x + yi$.

a. 1

b. $4 - 4i$

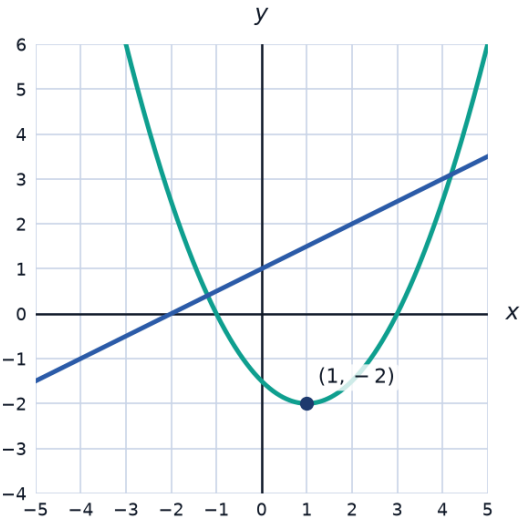
c. $3 - 2i$

d. $4 - 3i$

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39 [Challenging] • Target: Roots of Complex Numbers • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

For $x^2 - x - 56 = 0$, find the product of the roots.



a. -42

b. -47

c. -56

d. -51

40 [Challenging] • Target: Roots of Unity & Factorization • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

Simplify $(1 + 2i)(5 + 5i)$, giving your answer in the form $x + yi$.

a. $-3 + 16i$

b. $-5 + 15i$

c. $-1 + 12i$

d. $-3 + 14i$

41 [Challenging] • Target: Polar and Euler Forms • Cross-Lesson: 3.01 3D Vectors & the Cross Product

For $x^2 - 12x + 27 = 0$, find the product of the roots.

a. 35

b. 27

c. 39

d. 19

42 [Challenging] • Target: De Moivre's Theorem for Powers • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

For $x^2 - 11x + 18 = 0$, find the product of the roots.

a. 26

b. 19

c. 18

d. 25

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43 [Challenging] • Target: Roots of Complex Numbers • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

For $x^2 - 6x + 5 = 0$, find the sum of the roots.

a. 6

b. 11

c. 5

d. 3

44 [Challenging] • Target: Roots of Unity & Factorization • Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals

For $x^2 + 3x - 28 = 0$, find the sum of the roots.

a. -1

b. -3

c. -6

d. -7

45 [Challenging] • Target: Polar and Euler Forms • Cross-Lesson: 4.02 The Poisson Distribution

Write $z = 2(\cos 30^\circ + i\sin 30^\circ)$ in the form $x + yi$ (exact values).

a. $\sqrt{3} - i$

b. $\sqrt{3} + i$

c. $1 + \sqrt{3}i$

d. $-\sqrt{3} + i$

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